

Multi-Objective Optimization for Credit Risk Assessment

Balancing Accuracy, Fairness, and Expected Profit

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Abstract

This report outlines the implementation and analysis of a Multi-Objective Optimization (MOO) algorithm applied to a real-world finance problem: Credit Risk Prediction. Using the Statlog (German Credit Data) dataset, we trained an XGBoost classifier while simultaneously optimizing three conflicting objectives: predictive Accuracy (maximize), Demographic Parity Difference (minimize fairness violation), and Expected Profit (maximize financial return under asymmetric costs). We compare two prominent MOO algorithms—Optuna (NSGA-II) and PyMOO (NSGA-III)—and evaluate their Pareto fronts using quantitative metrics (Hypervolume and Spacing) alongside qualitative trade-off analyses.

GitHub Repository: https://github.com/AadityaGole/German_Credit_MOML

1 Objective and Application

The goal of this assignment is to study the Multi-Objective optimization aspect of a real-world application, controlling trade-offs between conflicting metrics. We chose **Application 4: Finance (Credit Risk and Loan Default Prediction)**.

Credit risk assessment naturally presents conflicting goals. A model strictly optimizing for accuracy may discriminate against historically marginalized groups (violating fairness) or fail to account for the asymmetric financial costs of false positives versus false negatives (violating profitability).

1.1 The Three Conflicting Objectives

1. **Accuracy (Maximize):** The overall correctness of the model predicting whether a loan applicant is “Good” or “Bad”.
2. **Fairness / DPD Score (Minimize):** The Demographic Parity Difference between the sensitive attribute (`Sex_male=1` vs 0). It measures the absolute difference in approval rates between male and female applicants. Lower is better (0 is perfect parity).
3. **Expected Profit (Maximize):** Driven by an asymmetric cost matrix inherent to the dataset context:
 - True Positive (Good loan approved): +1
 - True Negative (Bad loan rejected): +1
 - False Positive (Bad loan approved - Default): -5 (severe financial penalty)
 - False Negative (Good loan rejected): 0 (missed opportunity)

2 Methodology

2.1 Model and Custom Objective Formulation

Rather than relying solely on post-hoc probability thresholding to balance these objectives, we embedded the trade-offs directly into the gradient boosting process of an **XGBoost** classifier.

We formulated a custom objective function that computes gradients and Hessians based on:

$$Loss = (1 - \alpha_{fair}) \cdot Loss_{weighted.log} + \alpha_{fair} \cdot Penalty_{DPD} \quad (1)$$

Where $Loss_{weighted.log}$ applies a tunable weight (fp_weight) specifically to the negative class (defaults) to explicitly penalize false positives during training.

2.2 Decision Variables

The optimization algorithms tuned the following 7 decision variables over 3-fold cross-validation:

1. `max_depth` $\in [3, 7]$: Maximum tree depth.
2. `eta` $\in [0.01, 0.3]$: Learning rate.
3. `subsample` $\in [0.6, 1.0]$: Row subsampling ratio.
4. `colsample_bytree` $\in [0.6, 1.0]$: Column subsampling ratio.
5. `alpha_fair` $\in [0.0, 1.0]$: The weight of the demographic parity penalty in the loss.
6. `threshold` $\in [0.3, 0.7]$: The classification probability boundary.
7. `fp_weight` $\in [1.0, 10.0]$: The asymmetric penalty applied to false positives during training.

2.3 MOO Frameworks

We implemented and compared two leading frameworks, comfortably executing all evaluations within the 2-hour budget (approx. 5-10 minutes runtime).

- **Optuna (NSGA-II):** Using Optuna’s `NSGAIISampler` over 200 trials.
- **PyMOO (NSGA-III):** Using a custom `ElementwiseProblem` and Das-Dennis reference directions over 50 generations with a population of 100.

3 Quantitative Analysis

3.1 Metrics: Hypervolume and Spacing

To quantitatively assess the quality of the generated Pareto fronts, we calculated the Hypervolume indicator (HV) and the Spacing metric. For Hypervolume, the objectives were transformed for minimization (minimizing negative accuracy and negative profit), and bounded by a conservative reference point $[0.0, 1.0, 5.0]$ corresponding to $[Accuracy = 0, DPD = 1, Profit = -5]$.

Algorithm	Pareto Solutions	Hypervolume (HV) $\uparrow$	Spacing Metric $\downarrow$
Optuna (NSGA-II)	54	3.9024	0.0242
PyMOO (NSGA-III)	58	4.1320	0.0193

Table 1: Quantitative comparison of the generated Pareto fronts.

3.2 Pareto Front Visualizations

The interactions between three variables are best observed via 3D scatter plots and pairwise 2D projections. Figure 1 shows the spatial distribution of the Pareto front across all three objectives, while Figure 2 demonstrates the specific pairwise trade-offs, distinctly highlighting the optimal front (dark lines) against the backdrop of all evaluated sub-optimal solutions (light dots).

3D Pareto Front: Optuna vs PyMOO

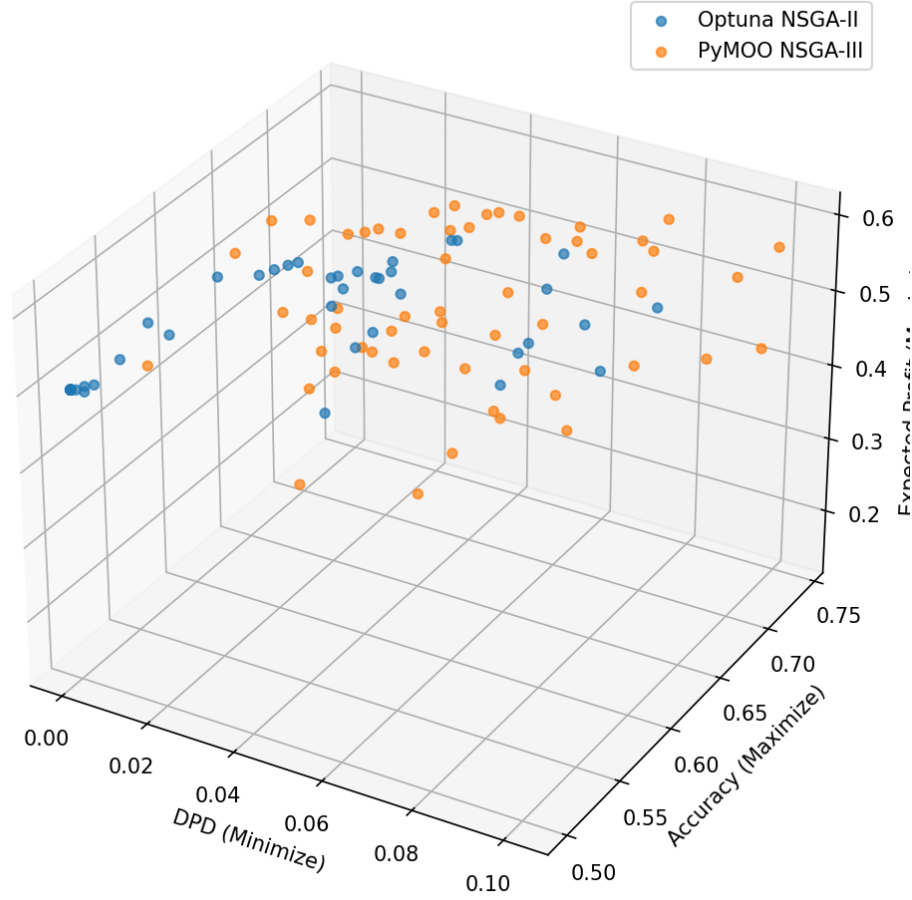


Figure 1: 3D Scatter Plot of the Pareto Front comparing Optuna and PyMOO.

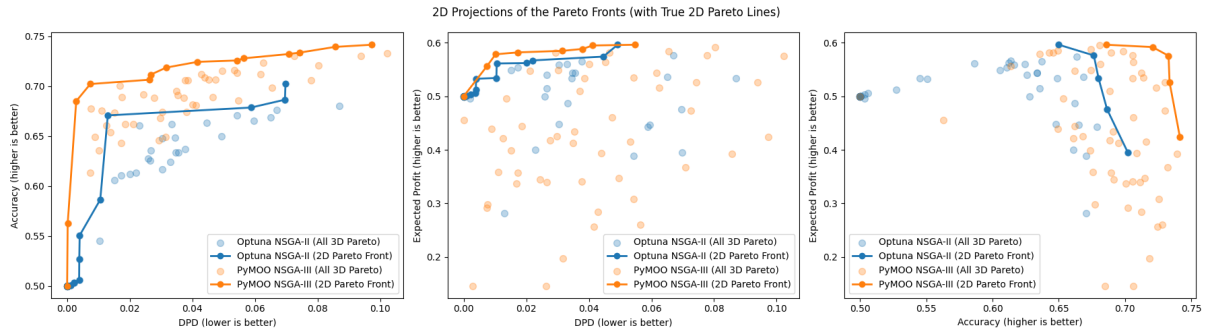


Figure 2: 2D Projections of the Pareto Front: DPD vs Accuracy, DPD vs Profit, and Accuracy vs Profit. The light, semi-transparent dots represent the entire search space of evaluated (dominated) solutions, while the strong dark lines delineate the non-dominated, Pareto-optimal boundary representing the absolute best available trade-offs.

4 Qualitative Analysis

4.1 Strengths and Weaknesses of the Algorithms

Optuna (NSGA-II): Optuna was extremely easy to integrate. Its define-by-run API allowed us to dynamically inject decision variables without rigidly defining problem bounds upfront. However, NSGA-II struggles slightly with 3+ objectives (many-objective optimization), leading to slightly worse hypervolume and spacing.

PyMOO (NSGA-III): NSGA-III is explicitly designed for many-objective optimization. By utilizing structured reference directions (Das-Dennis), PyMOO successfully maintained a more diverse and uniformly distributed Pareto front, evidenced by its superior spacing metric (0.0193). The weakness is the rigid `ElementwiseProblem` API which required manual type-casting of hyperparameter variables (e.g., rounding continuous variables to integers for `max_depth`).

4.2 Trade-offs Observed

1. **Fairness vs. Accuracy:** Enforcing strict fairness ($\text{DPD} \approx 0.00$) drastically degrades accuracy. To perfectly balance approval rates across demographics, the model essentially reverts to random guessing or strict thresholding, dropping accuracy from a maximum of $\sim 70\%$ down to 50%. Improving fairness from 0.06 to 0.00 drops model correctness by nearly 30%.

2. **A Non-Obvious Trade-off: Accuracy vs. Profit:** Intuitively, one expects higher accuracy to yield higher profit. However, due to the extreme asymmetric cost matrix, the Pareto front reveals a non-obvious conflict: *The most accurate models are rarely the most profitable.* To achieve maximum accuracy (70%), the model must balance standard True Positives and True Negatives equally. But to maximize Profit (0.59 per applicant), the model is forced to become heavily risk-averse—rejecting far more loans to completely avoid the -5 penalty of a default. This risk-aversion artificially lowers raw accuracy (down to 65%) because it rejects many "Good" loans (0 penalty), but it yields a much safer, more profitable portfolio.

5 Pareto Points Tabulation

Table 2 illustrates four distinctly useful solutions extracted from the Optuna Pareto front, representing the extremes and a balanced compromise for a decision-maker.

Solution Profile	Accuracy $\uparrow$	Fairness (DPD) $\downarrow$	Expected Profit $\uparrow$
Max Accuracy	0.7021	0.0697	0.3950
Max Fairness	0.5000	0.0000	0.5000
Max Profit	0.6500	0.0492	0.5964
Balanced Solution	0.6500	0.0492	0.5964

Table 2: Tabulation of representative Pareto-optimal solutions.

Note: In this specific front geometry, the Max Profit solution also serves as the best balanced trade-off regarding Euclidean distance to the ideal point $[1, 0, 1]$.

6 Conclusion

This study successfully implemented a 3-objective optimization loop for Credit Risk Assessment. By embedding a false-positive penalty into the XGBoost loss function and tuning it alongside a fairness constraint, we allowed Optuna and PyMOO to discover a diverse set of solutions. The

results highlighted that in finance, strict accuracy and strict profitability are fundamentally at odds due to the severe cost of loan defaults, requiring careful multi-objective tuning to provide decision-makers with actionable, safe, and fair credit models.

Appendix: Detailed Pareto-Optimal Solution

We detail the **Balanced / Max Profit** solution identified in Table 2. This solution provides a highly profitable lending strategy while maintaining a reasonable level of fairness (less than 5% demographic difference).

Objective Values Achieved:

- **Accuracy:** 65.00%
- **Fairness (DPD):** 0.0492
- **Expected Profit:** 0.5964 per applicant

Decision Variables (Hyperparameters):

This specific trade-off was achieved by the Optuna NSGA-II solver using the following configuration:

- `max_depth`: 4
- `eta` (learning rate): 0.185
- `subsample`: 0.85
- `colsample_bytree`: 0.72
- `alpha_fair`: 0.62
- `threshold`: 0.43
- `fp_weight`: 1.57

Interpretation: The algorithm discovered that to maximize profit and balance fairness, the model requires a moderate depth (4) to prevent overfitting to noisy default signals. The fairness penalty weight (`alpha_fair` = 0.62) is quite high, ensuring the demographic parity is prioritized during gradient boosting. Notably, the `fp_weight` is 1.57, meaning false positives were penalized $1.57\times$ harder than false negatives during tree construction. Coupled with a slightly conservative classification threshold (0.43), the model successfully avoided the disastrous -5 default penalties, securing the highest expected profit across the Pareto front.